

Random Variables: Discrete & Continuous

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Outline

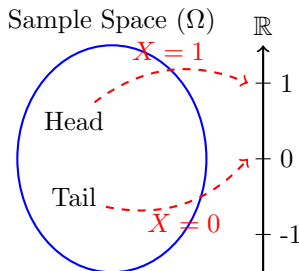
- 1 Random Variables
- 2 Distribution Functions: PMF & PDF
- 3 Expectation and Variance
- 4 Common Distributions

Chapter 1: Random Variables

Introduction to Random Variables

What is a Random Variable (RV)?

A **Random Variable** is a function that maps the outcome of a random process (uncertain event) to a real number. It translates “events” into “data”.



Types of Random Variables

Discrete Random Variable

- **Definition:** Takes on a countable number of distinct values (“jumps”).
- **Daily Life Example:** Number of students present in a class (you can’t have 2.5 students).
- **Industry Example:** Number of defective items in a batch of 100 manufactured parts.

Continuous Random Variable

- **Definition:** Takes on an infinite number of values within an interval (“smooth”).
- **Daily Life Example:** The exact time it takes to commute to work (e.g., 25.432 minutes).
- **Industry Example:** The exact weight of a package being shipped (e.g., 10.54 kg).

Numerical Example 1: The “Dice Sum”

Scenario: You roll two fair 6-sided dice. Let X be the random variable representing the **sum** of the two dice.

Task: List the possible values for X .

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Task: List the possible values for X .

- Minimum sum: $1 + 1 = 2$
- Maximum sum: $6 + 6 = 12$
- Range of X : $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Numerical Example 2: Defective Widgets

Scenario: A factory produces light bulbs. You pick 3 bulbs at random to test. Let X be the **number of defective bulbs** found.

Task: List possible values.

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Scenario: A factory produces light bulbs. You pick 3 bulbs at random to test. Let X be the **number of defective bulbs** found.

Task: List possible values.

- All bulbs work $\rightarrow X = 0$
- One is broken $\rightarrow X = 1$
- Two are broken $\rightarrow X = 2$
- All three broken $\rightarrow X = 3$

Sample Space Mapping: $\{GGG\} \rightarrow 0$, $\{BGG, GBG, GGB\} \rightarrow 1$, etc.

Quick Check: Random Variables

Quiz Q1: Discrete or Continuous?

Q1: Which of the following variables is Discrete?

- A) The time it takes for a website to load.
- B) The number of clicks on an advertisement.
- C) The temperature of the server room.
- D) The height of a user.

Quiz Q1: Discrete or Continuous?

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- A) The time it takes for a website to load.
- B) The number of clicks on an advertisement.
- C) The temperature of the server room.
- D) The height of a user.

Answer: B) The number of clicks Clicks are countable integers (0, 1, 2...). Time, temperature, and height are continuous.

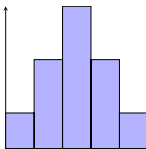
Chapter 2: PMF, PDF and CDF

Transition: Discrete to Continuous (Histograms to Curves)

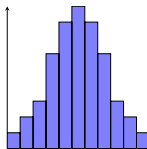
Imagine measuring the height of 10,000 people.

- As we increase the precision of our measurement (bins get narrower), the histogram approaches a smooth curve.
- In the limit (infinite precision), this curve becomes the **Probability Density Function (PDF)**.

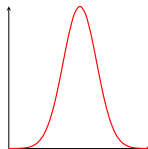
Coarse (Nearest 5cm)



Fine (Nearest 1mm)



Continuous (Infinite Precision)



Probability Mass Function (PMF)

The PMF, denoted as $P(X = x)$ or $p_X(x)$, gives the probability that the discrete random variable X is exactly equal to some value x .

Properties

- ① $0 \leq P(X = x) \leq 1$ for all x .
- ② $\sum_x P(X = x) = 1$ (The sum of all probabilities must be 1).

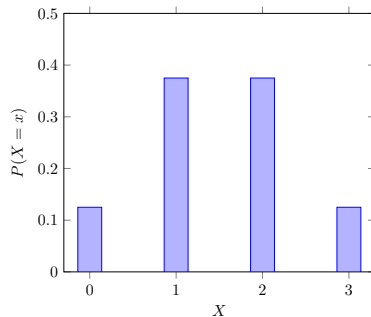
PMF Example 1: Load Balancing

Scenario: A server receives 3 requests. Each has a 50% chance of being “High Priority”. Let X = Number of High Priority requests.

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x	$P(X = x)$
0	$1/8 = 0.125$
1	$3/8 = 0.375$
2	$3/8 = 0.375$
3	$1/8 = 0.125$



PMF Example 3: Error Count

Scenario: A coder writes a function. The probability of having x bugs is given by $P(X = x) = k/2^x$ for $x \in \{1, 2, 3\}$, otherwise 0.

Find Constant k:

$$\sum P(X = x) = 1 \implies \frac{k}{2} + \frac{k}{4} + \frac{k}{8} = 1$$
$$\frac{4k + 2k + k}{8} = 1 \implies \frac{7k}{8} = 1 \implies k = 8/7$$

PMF:

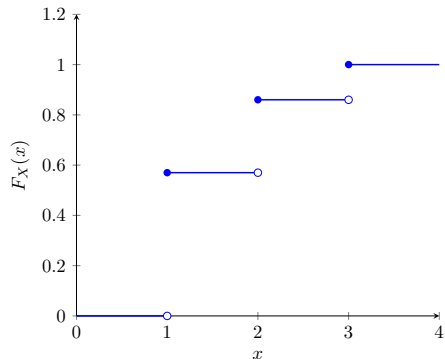
$$P(X = 1) = 4/7, \quad P(X = 2) = 2/7, \quad P(X = 3) = 1/7$$

Cumulative Distribution Function (CDF)

The CDF, $F_X(x) = P(X \leq x)$, represents the running total of probabilities. For discrete variables, it is a **Step Function**.

Using Example 3 (Errors):

- $x < 1$: 0
- $1 \leq x < 2$: $4/7 \approx 0.57$
- $2 \leq x < 3$:
 $4/7 + 2/7 = 6/7 \approx 0.86$
- $x \geq 3$: 1.0

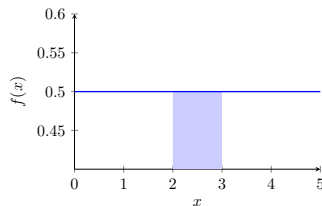


Probability Density Function (PDF)

For Continuous Random Variables, we use a **Probability Density Function (PDF)**, denoted $f(x)$.

Key Differences from PMF

- ❶ **PMF (Discrete):** $P(X = x)$ is the height of the bar.
- ❷ **PDF (Continuous):** The height $f(x)$ is **NOT** the probability.
- ❸ **Area is Probability:** Probability is the **Area under the curve**.



Calculating Probability (Integration)

To find the probability that X falls between a and b , we integrate the PDF:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties of PDF

- ① $f(x) \geq 0$ for all x . (Density cannot be negative).
- ② Total Area = 1: $\int_{-\infty}^{\infty} f(x) dx = 1$.

The Zero Probability Paradox

Concept: For a continuous variable, the probability of taking an **exact** value is Zero.

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Intuition: The Infinite Digital Scale Imagine a digital scale with infinite decimal places.

- You weigh an apple.
- Is it **exactly** 100g?
- It might be 100.000000001g or 99.999999998g.
- The probability of hitting exactly 100.000... (infinite zeros) is impossible (1 outcome out of infinite possibilities).

Conclusion: We only calculate probabilities for Intervals (e.g., between 99.0g and 101.0g).

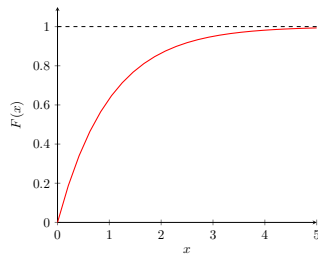
Cumulative Distribution Function (CDF)

The CDF, $F(x)$, gives the probability that the variable takes a value less than or equal to x .

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Relationship:

- PDF is the derivative of CDF:
 $f(x) = F'(x)$.
- Prob of interval:
 $P(a \leq X \leq b) = F(b) - F(a)$.



Numerical Example 1: The Linear PDF

Problem: Let X be a random variable with PDF:

$$f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(X > 0.5)$.

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Solution:

$$\begin{aligned} P(X > 0.5) &= \int_{0.5}^1 2x \, dx \\ &= [x^2]_{0.5}^1 \\ &= (1^2) - (0.5^2) = 1 - 0.25 = 0.75 \end{aligned}$$

Quick Check: PMF & PDF

Quiz Q1: Can Density Exceed 1?

Q1: Can the value of the Probability Density Function $f(x)$ be greater than 1?

- A) No, probability can never exceed 1.
- B) Yes, because $f(x)$ is density, not probability.

Quiz Q1: Can Density Exceed 1?

Q1: Can the value of the Probability Density Function $f(x)$ be greater than 1?

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- B) Yes, because $f(x)$ is density, not probability.

Answer: B. $f(x)$ represents density. For example, if $X \sim U(0, 0.1)$, the height is $1/(0.1) = 10$. The **Area** must be 1, but height can be anything.

Chapter 3: Expectation and Variance

One Set of Ideas, Two Kinds of Math

Expectation: The Weighted Average

The Expectation $E[X]$ is the **weighted average** of the possible values — the “center of gravity” of the distribution. The formula’s shape depends only on whether X is discrete or continuous:

Discrete: Sum

$$E[X] = \sum_x x \cdot P(X = x)$$

Continuous: Integral

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Same idea both times: multiply each value by how likely it is, then total it up. A sum just becomes an integral when there are infinitely many values to weigh.

Expectation: Worked Examples, Side by Side

Discrete: Fair Die Roll

$$E[X] = 1 \left(\frac{1}{6}\right) + 2 \left(\frac{1}{6}\right) + \cdots + 6 \left(\frac{1}{6}\right)$$

$$E[X] = \frac{21}{6} = 3.5$$

Continuous: $f(x) = 2x$ on $[0, 1]$

$$E[X] = \int_0^1 x \cdot (2x) dx = \int_0^1 2x^2 dx$$

$$E[X] = \left[\frac{2x^3}{3} \right]_0^1 = \frac{2}{3} \approx 0.67$$

Linearity of Expectation

Property (identical for discrete and continuous):

$$E[aX + b] = a E[X] + b$$

Real World Example: Project Cost Estimation

- Let X be the time (in hours) to complete a coding task. $E[X] = 10$ hours.
- A consultant charges \$50 per hour plus a \$100 fixed fee: $Y = 50X + 100$.

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Expected Cost: $50(10) + 100 = \$600$.

(We will revisit this example once Variance has been defined.)

Variance: Measuring the Spread

Variance measures how far values typically land from the mean $\mu = E[X]$:

Discrete: Sum

$$Var(X) = \sum_x (x - \mu)^2 P(X = x)$$

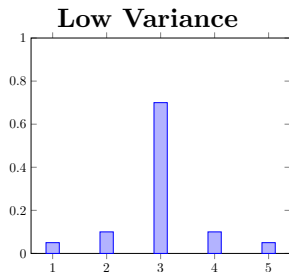
Continuous: Integral

$$Var(X) = \int (x - \mu)^2 f(x) dx$$

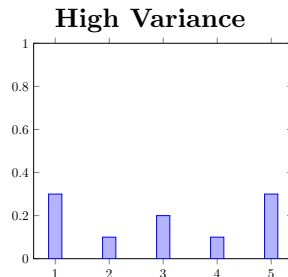
Standard Deviation: $\sigma_X = \sqrt{Var(X)}$ — same units as X itself.

High vs. Low Variance (Same Mean)

Both distributions below have the same Mean ($\mu = 3$), but different Variances.



Data is clustered near 3.



Data is spread far from 3.

The Shortcut Formula (Works for Both)

We use the property $Var(X) = E[X^2] - (E[X])^2$ for easier calculation. The proof only uses linearity of $E[\cdot]$, so it holds for discrete **and** continuous X unchanged.

- ① Let $\mu = E[X]$. By definition:

$$Var(X) = E[(X - \mu)^2]$$

- ② Expand the squared term:

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

- ③ Apply Linearity of Expectation:

$$E[X^2 - 2X\mu + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

- ④ Substitute $E[X] = \mu$:

$$\begin{aligned} &= E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 \implies E[X^2] - (E[X])^2 \end{aligned}$$

Variance: Worked Examples, Side by Side

Discrete: Daily Complaints

x	0	1	2
$P(X = x)$	0.2	0.5	0.3

$$\mu = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$$

$$\begin{aligned} \text{Var}(X) &= (0 - 1.1)^2(0.2) + (1 - 1.1)^2(0.5) \\ &\quad + (2 - 1.1)^2(0.3) \\ &= 0.242 + 0.005 + 0.243 = \mathbf{0.49} \end{aligned}$$

Continuous: $f(x) = 2x$ on $[0, 1]$ ($\mu = 2/3$, from earlier)

$$E[X^2] = \int_0^1 x^2(2x) dx = \left[\frac{2x^4}{4} \right]_0^1 = 0.5$$

$$\text{Var}(X) = 0.5 - \left(\frac{2}{3}\right)^2 = 0.5 - 0.444 = \mathbf{0.055}$$

Same shortcut formula, one uses a sum, the other an integral.

Variance Under Scaling (Works for Both)

Property: If $Y = aX + b$, then $Var(Y) = a^2 Var(X)$.

Proof:

- Let $\mu_X = E[X]$. Mean of Y is $\mu_Y = a\mu_X + b$.
- $Var(Y) = E[(Y - \mu_Y)^2] = E[((aX + b) - (a\mu_X + b))^2]$
- Simplify inside: $(aX - a\mu_X) = a(X - \mu_X)$.
- Square it: $a^2(X - \mu_X)^2$.
- Apply Expectation: $E[a^2(X - \mu_X)^2] = a^2 E[(X - \mu_X)^2] = a^2 Var(X)$.

Note: Adding a constant b shifts the distribution but does not change the spread.

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Note: Adding a constant b shifts the distribution but does not change the spread.

Back to Project Cost Estimation: recall $Var(X) = 4$ and $Y = 50X + 100$:

Variance of Cost: $50^2 \cdot Var(X) = 2500 \cdot 4 = 10,000$. **Std Dev:** $\sqrt{10,000} = \$100$.

Quick Check: Expectation & Variance

Quiz Q1: Expectation via Integration

Q1: Calculate $E[X]$ if $f(x) = 3x^2$ for $0 \leq x \leq 1$.

A) 0.5 B) 0.75 C) 1.0 D) 0.25

Quiz Q1: Expectation via Integration

Q1: Calculate $E[X]$ if $f(x) = 3x^2$ for $0 \leq x \leq 1$.

A) 0.5 B) 0.75 C) 1.0 D) 0.25

Answer: B) 0.75

$$E[X] = \int_0^1 x(3x^2)dx = \int_0^1 3x^3 dx = \left[\frac{3x^4}{4} \right]_0^1 = \frac{3}{4}$$

Quiz Q2: Variance Under Scaling

Q2: You are training a model. You have a random variable X with $Var(X) = 10$. You decide to normalize the data by doubling it and adding 5 ($Y = 2X + 5$). What is the variance of the new variable Y ?

- A) 25 B) 20 C) 40 D) 45

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A) 25 B) 20 C) 40 D) 45

Answer: C) 40

$$Var(aX + b) = a^2 Var(X)$$

$$Var(2X + 5) = 2^2 \cdot 10 = 4 \cdot 10 = 40$$

Quiz Q3: Standard Deviation from Moments

Q3: If $E[X] = 5$ and $E[X^2] = 30$, what is the Standard Deviation σ ?

- A) 5 B) $\sqrt{5}$ C) 25 D) $\sqrt{30}$

Quiz Q3: Standard Deviation from Moments

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A) 5 B) $\sqrt{5}$ C) 25 D) $\sqrt{30}$

Answer: B) $\sqrt{5}$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 30 - 25 = 5 \implies \sigma = \sqrt{5}$$

Chapter 4: Common Distributions

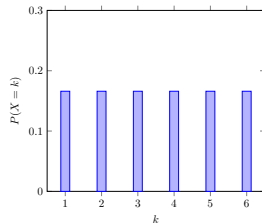
Discrete and Continuous

1. Uniform Distribution: Discrete vs Continuous

Same idea both times: every outcome in the range is equally likely.

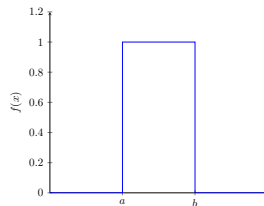
Discrete: $X \sim U(a, b)$

- $n = b - a + 1$ integer outcomes
- $P(X = k) = \frac{1}{n}$
- Mean: $\frac{a + b}{2}$
- Var: $\frac{n^2 - 1}{12}$



Continuous: $X \sim U(a, b)$

- Any real value in $[a, b]$
- $f(x) = \frac{1}{b - a}$
- Mean: $\frac{a + b}{2}$
- Var: $\frac{(b - a)^2}{12}$



2. Bernoulli Distribution

Definition: Single trial with binary outcome (0 or 1).

Use Cases:

- Tossing a coin
- Pass/Fail in an exam.
- an item being defective or not.

Notation: $X \sim \text{Bern}(p)$

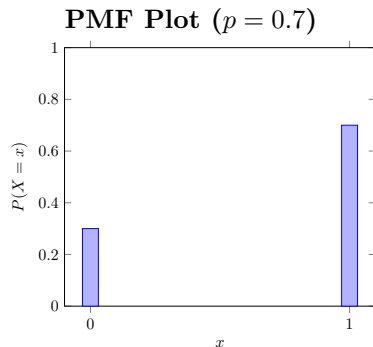
Parameter: p (prob of success).

PMF:

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Properties:

- **Mean:** $E[X] = p$
- **Variance:** $\text{Var}(X) = p(1 - p)$



3. Binomial Distribution

Definition: Number of successes in n independent Bernoulli trials.

Use Cases:

- **Quality Control:** If you test 20 items (n), and each has 10% chance of defect (p), what is the probability that less than 5 items are defective?

Notation: $X \sim \text{Bin}(n, p)$

Parameters:

- n : The total number of independent trials.
- p : The probability of success in a single trial.

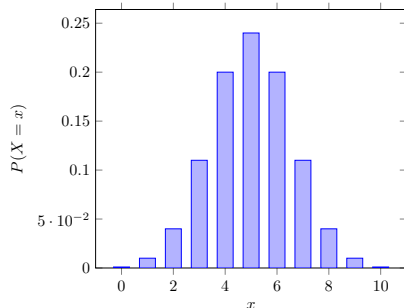
PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad ; \quad 0 \leq k \leq n$$

Properties:

- **Mean:** $E[X] = np$
- **Variance:** $\text{Var}(X) = np(1 - p)$

PMF ($n = 10, p = 0.5$)



4. Poisson Distribution

Definition: Models the number of events occurring in a fixed interval of time or space.

Use Cases:

- **Queues:** the probability of receiving a specific number of customer service calls during a peak hour.
- **Website Traffic:** Predicting the number of visitors or server requests per minute.

Notation: $X \sim Pois(\lambda)$

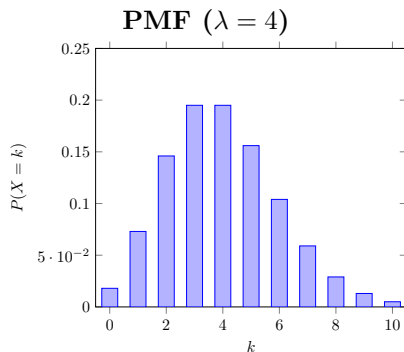
Parameter: λ : The average number of events per interval.

PMF:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Properties:

- **Mean:** $E[X] = \lambda$
- **Variance:** $Var(X) = \lambda$



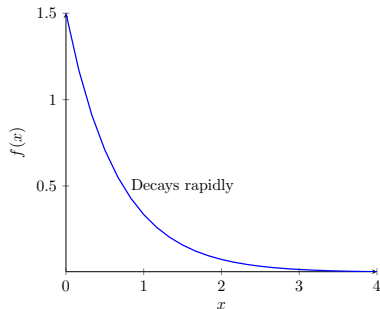
5. Exponential Distribution

Notation: $X \sim \text{Exp}(\lambda)$

Use Case: Modeling time until an event (e.g., time until next customer arrives).

Parameter: λ : commonly known as the rate parameter.

- **PDF:** $f(x) = \lambda e^{-\lambda x}$ ($x \geq 0$)
- **Mean:** $1/\lambda$
- **Variance:** $1/\lambda^2$



Exponential: Memoryless Property

The Property

$$P(X > s + t \mid X > s) = P(X > t)$$

Interpretation: The probability of waiting t more minutes is the same, regardless of how long (s) you have already waited.

Real Life Example:

- **Radioactive Decay:** An atom doesn't "age". The chance it decays in the next second is constant, whether it's 1 second old or 1 billion years old.
- **Customer Arrivals:** If arrivals are random (Poisson process), the time you've already waited for a customer does not change how much longer you will likely wait.

6. Normal (Gaussian) Distribution

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

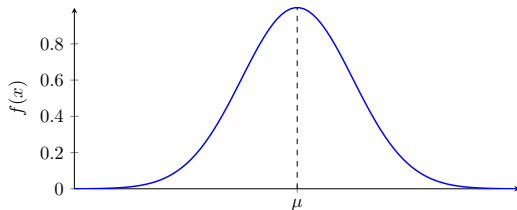
Description: The “Bell Curve”. Most common distribution in nature.

Parameters:

- μ (Mean/Location): This represents the center or peak of the “bell”
- σ^2 (Variance/Scale): This represents the “spread” of the distribution.

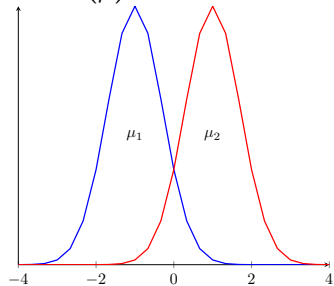
PDF:

$$f(x | \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

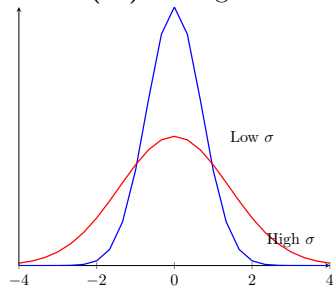


Normal Distribution: Parameters

Mean (μ): shifts the center.



Variance (σ^2): changes the width.

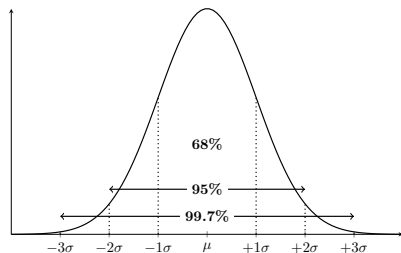


The 68-95-99.7 Rule (Empirical Rule)

For a Normal Distribution:

- **68%** of the data falls within **one** standard deviation of the mean ($\mu \pm 1\sigma$).
- **95%** of the data falls within **two** standard deviations ($\mu \pm 2\sigma$).
- **99.7%** of the data falls within **three** standard deviations ($\mu \pm 3\sigma$).

Interpretation: It is extremely rare (only 0.3% chance) for data to fall outside the 3σ range.



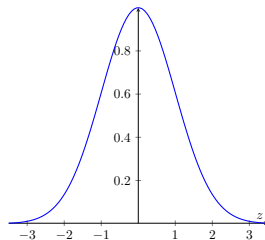
Standard Normal Distribution (Z)

If we scale any Normal variable, we get the **Standard Normal**:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Properties:

- Mean $\mu = 0$
- Std Dev $\sigma = 1$
- PDF: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$
- Used for “Z-Scores” (how many sigmas away is a data point?).



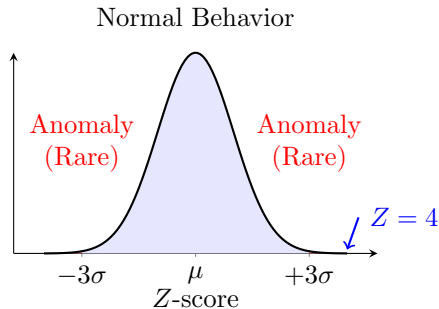
Anomaly Detection using Standard Normal Distribution

Concept:

- In a Normal distribution, values far from the mean (μ) are rare.
- We calculate the **Z-score**: $Z = \frac{x - \mu}{\sigma}$.
- **Threshold**: Typically, if $|Z| > 3$ (more than 3 standard deviations away), the data point is flagged as an **Anomaly** or **Outlier**.

Example: Network Security

- Normal traffic: $\mu = 50$ Mbps, $\sigma = 5$ Mbps.
- **Event**: Current traffic is 70 Mbps.
- $Z = \frac{70 - 50}{5} = 4$.
- Since $4 > 3$, this is an **Anomaly** (Potential DDoS attack).



Numerical Problem 1: Manufacturing

Scenario: A machine cuts rods. Length $L \sim \mathcal{N}(10\text{cm}, 0.04\text{cm}^2)$ ($\sigma = 0.2$). Rods must be between 9.6 and 10.4. What % are defective?

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Solution:

- ① Limits are 10 ± 0.4 .
- ② Since $\sigma = 0.2$, the limits are exactly $\mu \pm 2\sigma$.
- ③ We know 95% are inside $\pm 2\sigma$.
- ④ Therefore, $100\% - 95\% = 5\%$ are defective.

Numerical Problem 2: Credit Card Fraud Detection

Scenario: A bank's fraud detection system monitors the spending habits of a customer. Their transaction amounts follow a Normal Distribution with a mean $\mu = \$50$ and a standard deviation $\sigma = \$15$.

The system is programmed to flag any transaction with a Z-score greater than 3 ($|Z| > 3$) as “**Suspicious Activity**”.

Question: A new transaction of \$100 is attempted on the card. Will the system flag this transaction?

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① Calculate the Z-score:

$$Z = \frac{x - \mu}{\sigma} = \frac{100 - 50}{15} = \frac{50}{15} \approx 3.33$$

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Solution:

- 1 Calculate the Z-score:

$$Z = \frac{x - \mu}{\sigma} = \frac{100 - 50}{15} = \frac{50}{15} \approx 3.33$$

- 2 **Compare with Threshold:** Since $3.33 > 3$, the transaction is more than 3 standard deviations above the average spending.
- 3 **Conclusion:** The system **will flag** this transaction as suspicious.

Numerical Problem 3: Inverse Problem

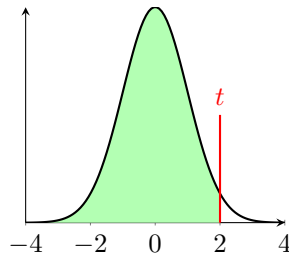
Scenario: Delivery time $T \sim \mathcal{N}(30, 16)$ ($\sigma = 4$). We want to promise a time t such that we are 97.5% sure we arrive before t .

Numerical Problem 3: Inverse Problem

Scenario: Delivery time $T \sim \mathcal{N}(30, 16)$ ($\sigma = 4$). We want to promise a time t such that we are 97.5% sure we arrive before t .

Solution:

- We need the cumulative area to be 0.975.
- 95% is middle. The tails are 2.5% each.
- Area to left of $+2\sigma$ is
50%(left half) + 47.5%(half of 95) = 97.5%.
- $t = \mu + 2\sigma = 30 + 2(4) = 38$ mins.



Quick Check: Distributions

Quiz Q1: Poisson Probability

Q1: A call center receives an average of $\lambda = 3$ calls per minute (Poisson). What is the probability of receiving exactly 0 calls in a minute?

A) 0 B) e^{-3} C) $3e^{-3}$ D) 0.5

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Answer: B) e^{-3}

$$P(X = 0) = \frac{e^{-3} \cdot 3^0}{0!} = e^{-3}$$

Quiz Q2: Identify the Distribution

Q2: We perform 5 independent trials. Each trial has a success probability of 0.2. Which distribution models the number of successes?

- A) Bernoulli(0.2)
- B) Poisson(5)
- C) Binomial(5, 0.2)
- D) Uniform

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Answer: C) Binomial(5, 0.2) Because we are counting successes in a fixed number ($n = 5$) of independent trials.

Quiz Q3: Normal Tail Probability

Q3: For a Normal distribution, what is the probability of being more than 3 standard deviations away from the mean (in either direction)?

A) 5% B) 1% C) 0.3% D) 0.0%

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Answer: C) 0.3% 99.7% is inside, so $100 - 99.7 = 0.3\%$ is outside.

Summary

- **Random Variables** map real-world uncertainty to numbers — **discrete** (countable jumps) or **continuous** (a smooth interval).
- **PMF** gives the probability of an exact value (discrete); **PDF** gives density, where **area** is probability (continuous). Both have a **CDF**: the running total $P(X \leq x)$.
- **Expectation** is the center; **Variance** is the spread — computed by a sum for discrete X , an integral for continuous X , using exactly the same properties either way.
- **Key Discrete Distributions**: Uniform, Bernoulli (1 trial), Binomial (n trials), Poisson (rates/queues).
- **Key Continuous Distributions**: Uniform, Exponential (memoryless waiting times), Normal (the bell curve, central to statistics).
- **68-95-99.7 Rule**: Quick estimation for Normally distributed data, and the basis of Z-score anomaly detection.

Thank You!